

Uncertainty Quantification for Domain Agnostic Multi-Agent Workflows for Scientific Visualization

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Introduction

THE PROBLEM

Uncertainty can accumulate across an agentic workflow, negatively impacting the accuracy of the final output. Bringing in a human to provide feedback can reduce this accumulation but requires knowing when and where feedback is most effective.

How do we determine where to place human checkpoints in a multi-agent workflow?

THE SYSTEM

AEVIS (Automatic Exploratory VISualization) is a domain-agnostic, multi-agent workflow for scientific visualization. It follows a predetermined plan to generate a Cinema image database from an input dataset. Uncertainty in earlier steps (like selecting the wrong variables to visualize) reduces the accuracy and usefulness of the output visualization images.

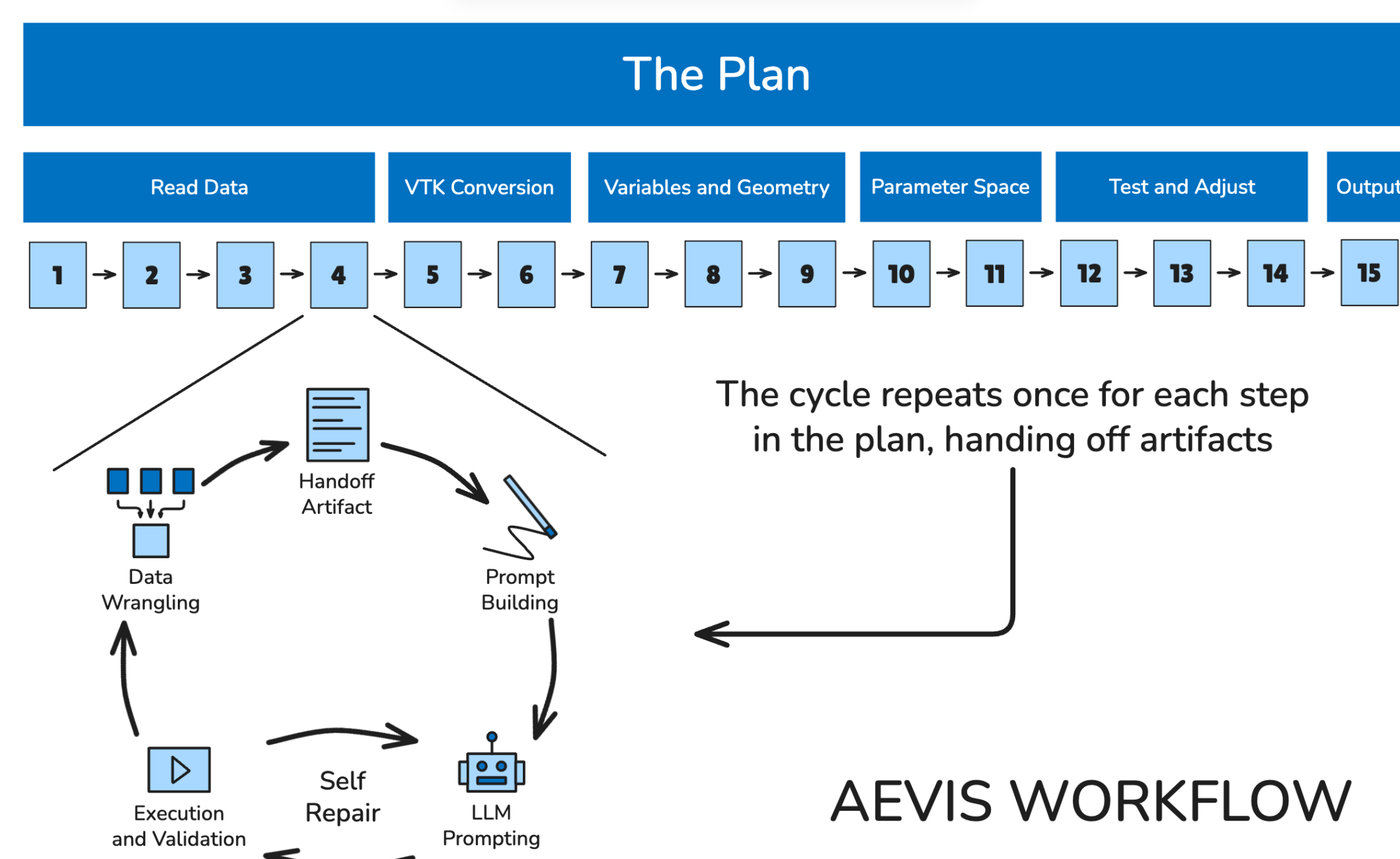


Figure 1. The AEVIS workflow.

THE SOLUTION

We propose using uncertainty quantification (UQ) to determine where to place human checkpoints. Our UQ measurement technique is to use multiple rounds generation and then compare the semantic similarity between the outputs from each round.

Methods

THE PROCESS

We ran an experiment using our proposed UQ method to determine the uncertainty contribution of different workflow steps. We ran the workflow 50 times on each dataset (Wildfire and NIF flash) and extracted the explanations given by the LLM for each identified variable in the given dataset. We then used a general, pretrained BERT model to get the semantic similarities of each combination of these explanations for each variable across runs.

THE CONDITIONS

We varied the restart point used in each set of runs to control what was held constant between runs. This enabled evaluating the effect of variation in different workflow steps on uncertainty.

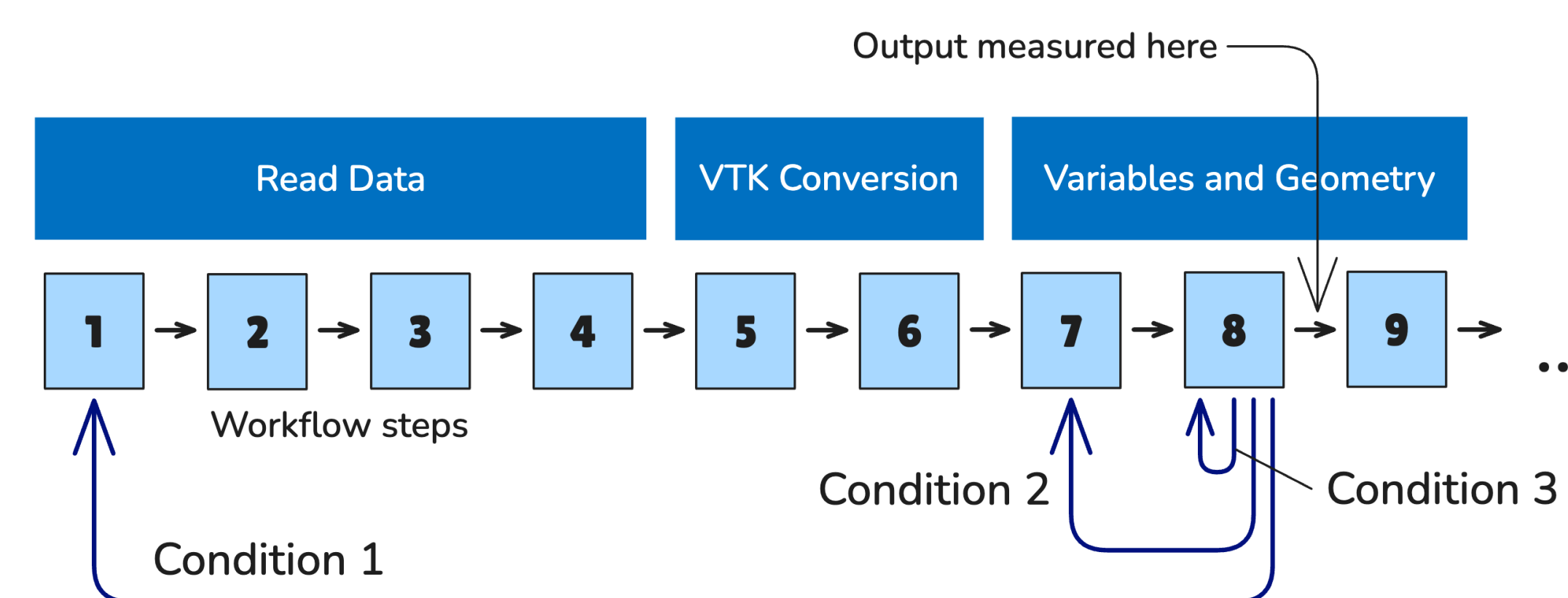


Figure 2. The AEVIS workflow steps with the repeat conditions shown.

Results

THE EFFECT

Repeating multiple steps in each round changed the semantic similarity relative to repeating only a single step. How many steps were repeated, however, did not have a significant effect. The direction of this change varied for different variable types.

KEY IDEA

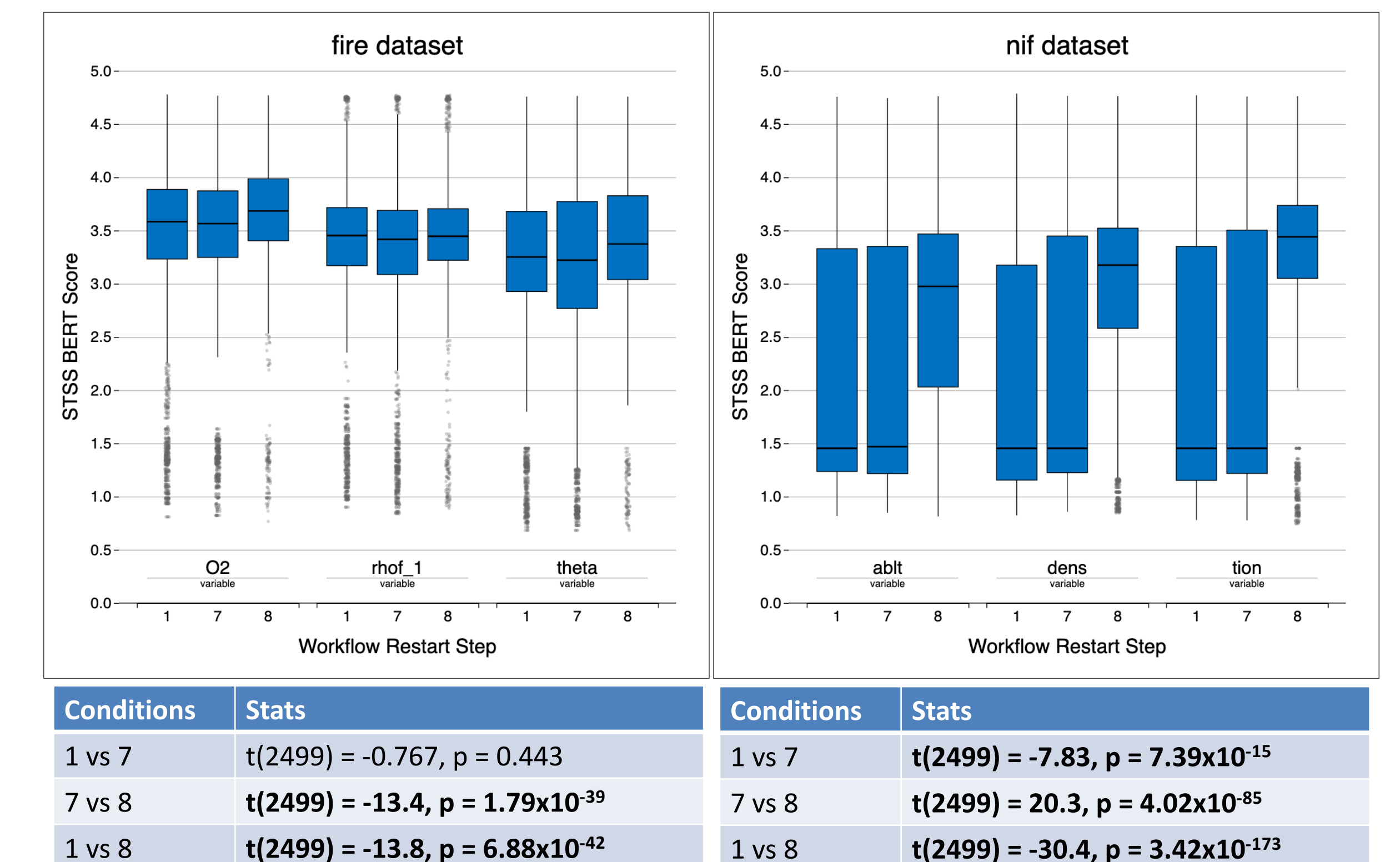


Figure 3. Semantic Similarity of LLM explanations for identified variables.

Discussion

THE IMPLICATION

The insignificant effect of the number of steps repeated shows there is a nonlinear relationship between uncertainty and how many steps are repeated in each round. This suggests blocks of steps can be treated as a single unit for uncertainty contribution. Uncertainty jumps after each block thus map out potential locations for placing human checkpoints in the workflow.

THE CONFOUND

The variation in this pattern was greater for the more complex dataset (NIF flash), and for some variable types (like coordinate). Additional work is needed to quantify these effects and determine if interactions between specific variable types or data formats and specific workflow steps cause the difference.

Conclusion

THE IMPACT

We have demonstrated a UQ measurement technique for multi-agent workflows that can be used to determine the contribution of prior steps to the uncertainty in the output of a future step. Future work will include using this technique to fully map out such contributions across the workflow. This technique provides a reproduceable, systematic way to select human checkpoints.